ANSWER SCHEME TRIAL PSPM SEM 1 2025/2026

NO		ANSWER SCHEME	MARKS	REMARKS
SECTION A				
1.	(a)	$=\lim_{p\to 2}\frac{(p-2)(p+2)}{(2-p)(4+2p+p^2)}$	**K1**	Factorise
		$=\lim_{p\to 2}\frac{(p-2)(p+2)}{-(p-2)(4+2p+p^2)}$	**K1**	Simplify
		$=\lim_{p\to 2}-\frac{(p+2)}{(4+2p+p^2)}$	**J1**	
		$=-\frac{(2+2)}{(4+2(2)+(2)^2)}$	**K1√**	Substitute
		$=-\frac{1}{3}$	**J1**	Answer
	(b)	$=\lim_{x\to 4}\frac{x-4}{\sqrt{2x-1}-\sqrt{x+3}}\times\frac{\sqrt{2x-1}+\sqrt{x+3}}{\sqrt{2x-1}+\sqrt{x+3}}$	**K1**	Multiplication with correct conjugate
		$=\lim_{x\to 4}\frac{(x-4)(\sqrt{2x-1}+\sqrt{x+3})}{(2x-1)-(x+3)}$		
		$=\lim_{x\to 4}\frac{(x-4)(\sqrt{2x-1}+\sqrt{x+3})}{x-4}$	**J1**	
		$=(\sqrt{2(4)-1}+\sqrt{4+3})$	**K1√**	Substitute
		$=2\sqrt{7}$	**J1**	Answer
		TOTAL	**9 M**	
2.	(a)	$=\lim_{h\to 0}\frac{2\sqrt{x+h}-2\sqrt{x}}{h}$	**B1**	First principle
		$=\lim_{h\to 0}\frac{2\sqrt{x+h}-2\sqrt{x}}{h}\times\frac{2\sqrt{x+h}+2\sqrt{x}}{2\sqrt{x+h}+2\sqrt{x}}$	**K1**	Multiplication with correct conjugate and solving
		$=\lim_{h\to 0}\frac{4(x+h-x)}{h\left[2\sqrt{x+h}+2\sqrt{x}\right]}$		
		$=\lim_{h\to 0}\frac{4}{\left[2\sqrt{x+h}+2\sqrt{x}\right]}$	**J1**	Correct term
		$=\frac{4}{\left[2\sqrt{x+0}+2\sqrt{x}\right]}$	**K1√**	Substitute $h=0$
		$=\frac{4}{4\sqrt{x}}$		
		$=\frac{1}{\sqrt{x}}$	**J1**	

	(b)	$y = \ln\left(x^2\sqrt{1-3x}\right)$ $y = \ln(x^2) + \ln(1-3x)^{\frac{1}{2}}$ $y = 2\ln x + \frac{1}{2}\ln(1-3x)$ $\frac{dy}{dx} = 2\left(\frac{1}{x}\right) + \frac{1}{2}\left(\frac{-3}{1-3x}\right)$ $\frac{dy}{dx} = \frac{2}{x} - \frac{3}{2(1-3x)}$ ALTERNATIVE $y = \ln\left(x^2\sqrt{1-3x}\right)$ $\frac{dy}{dx} = \frac{1}{x^2\sqrt{1-3x}}\frac{d}{dx}\left(x^2\sqrt{1-3x}\right)$ $\frac{dy}{dx} = \frac{1}{x^2\sqrt{1-3x}}\left[x^2.\frac{d}{dx}\left(\sqrt{1-3x}\right) + \sqrt{1-3x}.\frac{d}{dx}(x^2)\right]$ $\frac{dy}{dx} = \frac{1}{x^2\sqrt{1-3x}}\left[x^2\left(\frac{-3}{2\sqrt{1-3x}}\right) + 2x\sqrt{1-3x}\right]$ $\frac{dy}{dx} = -\frac{3}{2(1-3x)} + \frac{2}{x}$	 **K1** **J1** **K1** **J1**	
		TOTAL	**9 M**	
3.		$\frac{dV}{dt} = -6\ \text{cm}^3/\text{s}$ $\frac{r}{h} = \frac{4}{8} \Rightarrow r = \frac{h}{2}$ $V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$ $\frac{dV}{dh} = \frac{\pi h^2}{4}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{4}{\pi h^2} \times (-6)$ $= \frac{-24}{\pi h^2}$ $h = 3 \Rightarrow \frac{dh}{dt} = \frac{-24}{\pi(3)^2}$ $= -\frac{8}{3\pi}$ cm/s	 **B1** **K1** **K1** **K1** **J1** **K1** **J1**	 Correct definition in volume Diff correctly Find $\frac{dh}{dt}$ Substitute $h = 3$
		TOTAL	**7 M**	

SECTION B				
1.	$\frac{3+2i}{(1-4i)i^3}=\frac{3+2i}{(1-4i)(-i)}=\frac{3+2i}{-i+4i^2}$		**K1**	$i^3=-i$
	$=\frac{3+2i}{-4-i}$			
	$=\frac{3+2i}{-4-i}\times\frac{-4+i}{-4+i}$		**K1√**	Correct conjugate
	$=\frac{-12+3i-8i+2i^2}{16-i^2}$			
	$=\frac{-12-5i-2}{17}$			
	$=-\frac{14}{17}-\frac{5}{17}i$		**J1**	In term of $a+bi$
	$\left\lvert\frac{3+2i}{(1-4i)i^3}\right\rvert=\sqrt{\left(-\frac{14}{17}\right)^2+\left(-\frac{5}{17}\right)^2}$		**K1√**	Correct formulae
	$=\frac{\sqrt{221}}{17}$		**J1**	Exact value
		TOTAL	**5 M**	
2.	(a)	$\log_a\left(\frac{x}{a^2}\right)=3\log_a 2-\log_a(x-2a)$		
		$\log_a\left(\frac{x}{a^2}\right)=\log_a 2^3-\log_a(x-2a)$		
		$\log_a\left(\frac{x}{a^2}\right)=\log_a\left(\frac{8}{x-2a}\right)$	**K1**	Apply law
		$\left(\frac{x}{a^2}\right)=\left(\frac{8}{x-2a}\right)$	**K1**	Compare
		$x^2-2ax=8a^2$		
		$x^2-2ax-8a^2=0$	**K1**	Quadratic equation
		$(x-4a)(x+2a)=0$	**J1**	Both *x*
		$x=4a$ or $x=-2a$		
		since $a>0, x\neq-2a$		
		$\therefore x=4a$	**J1**	*x* in term of a
	(b)	$\frac{24-8x}{x^2-9}\geq-2$		
		$\frac{24-8x}{x^2-9}+2\geq0$	**K1**	Equalize denominator
		$\frac{24-8x+2x^2-18}{x^2-9}\geq0$		
		$\frac{2x^2-8x+6}{x^2-9}\geq0$	**K1**	Simplify
		$\frac{2(x-1)(x-3)}{(x-3)(x+3)}\geq0, x\neq-3,3$	**J1**	

-3 1 3

x	$(-\infty,-3)$	$(-3,1]$	$[1,3)$	$(3,\infty)$
$x-1$	-	-	+	+
$x-3$	-	-	-	+
$x-3$	-	-	-	+
$x+3$	-	+	+	+
Conclusion	+	-	+	+

K1√ Attempt to find solution set

K1√ Attempt to find positive solution

$\therefore (-\infty,-3)\cup[1,3)\cup(3,\infty) / \{x : x<-3 \text{ or } x\geq 1 \setminus x\neq 3\}$ **J1**

ALTERNATIVE

$$\frac{24-8x}{x^2-9}\geq -2$$

$$\frac{24-8x}{x^2-9}+2\geq 0$$

$$\frac{24-8x+2x^2-18}{x^2-9}\geq 0$$

$$\frac{2x^2-8x+6}{x^2-9}\geq 0$$

$$\frac{2(x-1)(x-3)}{(x-3)(x+3)}\geq 0$$

$$\frac{2(x-1)}{(x+3)}\geq 0,\quad x\neq 3$$

-3 1

x	$(-\infty,-3)$	$(-3,1]$	$[1,\infty)$
$x-1$	-	-	+
$x+3$	-	+	+
Conclusion	+	-	+

$\therefore (-\infty,-3)\cup[1,3)\cup(3,\infty) / \{x : x<-3 \text{ or } x\geq 1 \setminus x\neq 3\}$

TOTAL	**11 M**

<table>
<tr><td>3.</td><td>(a)</td><td>$f \circ g(x) = \sqrt{4x-1}$
$f(e^{x+2}) = \sqrt{4x-1}$
$g\left(g^{-1}(x)\right) = x$
$e^{g^{-1}(x)+2} = x$
$g^{-1}(x) = \ln x - 2$
$f(x) = \sqrt{4(\ln x - 2) - 1}$
$= \sqrt{4\ln x - 9}$

$4\ln x - 9 \geq 0$
$\ln x \geq \frac{9}{4}$
$x \geq e^{\frac{9}{4}}$
$D_{f(x)} : \left[e^{\frac{9}{4}}, \infty\right)$

ALTERNATIVE
$f \circ g(x) = \sqrt{4x-1}$
$f(e^{x+2}) = \sqrt{4x-1}$
Let $u = e^{x+2}$
$x = \ln u - 2$
$f(u) = \sqrt{4(\ln u - 2) - 1} = \sqrt{4\ln u - 9}$
$f(x) = \sqrt{4\ln x - 9}$</td><td>K1

K1
J1

K1√

J1</td><td>Find $g^{-1}(x)$

Correct $f(x)$

Find domain</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td></td><td>(b)</td><td>$(g \circ f)(e^3) = e^{\sqrt{4\ln e^3 - 9}+2}$
$= e^{\sqrt{3}+2}$</td><td>K1√

J1</td><td>$(g \circ f)(e^3)$</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td colspan="3">TOTAL</td><td>7 M</td><td></td></tr>
<tr><td colspan="5"></td></tr>
<tr><td>4.</td><td>(a)</td><td>$f(x) > 1$
$\frac{3}{1-x} > 1$
$\frac{3}{1-x} - 1 > 0$
$\frac{3}{1-x} - \frac{1-x}{1-x} > 0$
$\frac{x+2}{1-x} > 0$

<table><tr><td>$x+2$</td><td>$-$</td><td>$+$</td><td>$+$</td></tr><tr><td>$1-x$</td><td>$+$</td><td>$-$</td><td>$-$</td></tr><tr><td>$\frac{x+2}{1-x} > 0$</td><td>$-$</td><td>$+$</td><td>$-$</td></tr></table>
$\{x : -2 < x < 1\}, \cap\ x > 1 \Rightarrow \{\ \}$ / no solution</td><td>K1

J1</td><td>Simplify and get rational inequality

Answer</td></tr>
<tr><td colspan="5"></td></tr>
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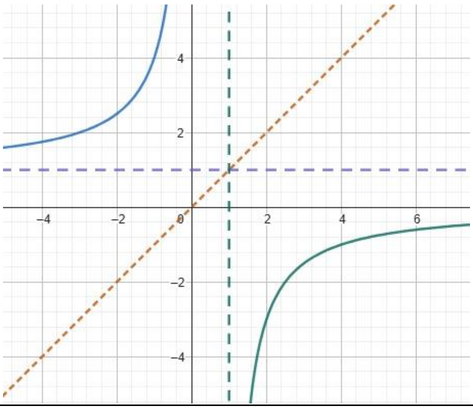
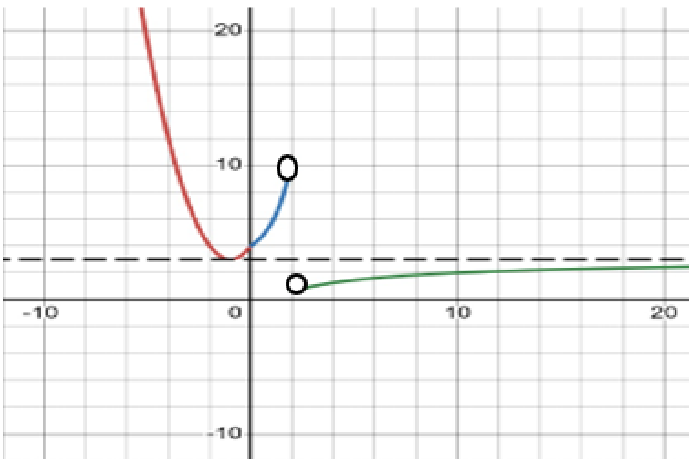
	(b)	$f(x_1) = f(x_2)$ $\dfrac{3}{1-x_1} = \dfrac{3}{1-x_2}$ $1 - x_1 = 1 - x_2$ $\therefore\ x_1 = x_2$ Therefore $f(x)$ is one-to-one function	**B1** **K1** **J1**	Definition and correct function Simplify Conclusion
	(c)	$f\left(f^{-1}(x)\right) = x$ $\dfrac{3}{1-f^{-1}(x)} = x$ $1 - f^{-1}(x) = \dfrac{3}{x}$ $f^{-1}(x) = 1 - \dfrac{3}{x}$ $f^{-1}(x) = \dfrac{x-3}{x} = 1 - \dfrac{3}{x}$	**K1** **K1** **J1**	Using properties of inverse (composite function) Solving Answer
	(d)	$D_{f^{-1}(x)} = (-\infty, 0)$ $R_{f^{-1}(x)} = (1, \infty)$	**B1** **B1**	
	(e)		**R1** **R1** **R1**	Graph shape All asymptote corrects All correct included label and intercepts.
		TOTAL	**13 M**	
5.	(a)	$\lim\limits_{x\to 0^+} e^x + 3m = 1 + 3m$ $\lim\limits_{x\to 0^-} x^2 + 2x - 5m + 9 = -5m + 9$ $w(0) = \lim\limits_{x\to 0^-} w(x) = \lim\limits_{x\to 0^+} w(x)$ $e^0 + 3m = -5m + 9 = 1 + 3m$ $1 + 3m = -5 + 9$ $m = 1$	**K1** **J1** **B1** **K1√** **J1**	Find limit of 0 from right & left Both terms correct Find m Answer
	(b)	Since $w(x)$ is not defined at $x = 2$, therefore $w(x)$ is not continuous at $x = 2$	**K1√J1**	Answer and reason

		OR $e^2+3(1)\neq\frac{3(2)-2}{2+4}$, therefore $w(x)$ is continuous at $x=2$		
	(c)	for $x>2$, $\quad f(x)=\frac{3x-2}{x+4}, x\neq -4$ $\lim\limits_{x\to+\infty}\frac{3x-2}{x+4}=\lim\limits_{x\to+\infty}\frac{\frac{3x}{x}-\frac{2}{x}}{\frac{x}{x}+\frac{4}{x}}$ $=\lim\limits_{x\to+\infty}\frac{3-\frac{2}{x}}{1+\frac{4}{x}}$ $=\frac{3-0}{1+0}$ $=3$ Thus, $y=3$ is horizontal asymptote.	K1 K1 J1	Find lim at ∞ and divide top and bottom with x Attempt to find lim at ∞
	(d)	 Range $=\left(\frac{2}{3},\infty\right)$	R1 R1 R1 B1	Correct shape parabola Correct shape for $x>2$ & two o (empty circle) All correct with label
		TOTAL	**14 M**	
6.	(a)	$y^2sinx=cosy^3$ $u=y^2 \quad v=sinx$ $u'=2y \quad v'=cosx\frac{dx}{dy}$ $2ysinx+y^2cosx\frac{dx}{dy}=-siny^3.3y^2$ $y^2cos\,x\frac{dx}{dy}=-3y^2siny^3-2ysinx$ $\frac{dx}{dy}=\frac{-(3y^2siny^3+2ysinx)}{y^2cos\,x}$ $=\frac{-y(3y\ \ siny^3+2sinx)}{y^2cos\,x}$ $=\frac{-(3y\ \ siny^3+2sinx)}{y\ \ cos\,x}$	K1 K1 √JI K1 K1 J1	Product rule and implicit Diff cos x

		ALTERNATIVE $2y\sin x\dfrac{dy}{dx}+y^2\cos x=-3y^2\sin y^3\dfrac{dy}{dx}$ $(3y^2\sin y^3+2y\sin x)\dfrac{dy}{dx}=-y^2\cos x$ $\dfrac{dy}{dx}=\dfrac{-y^2\cos x}{3y^2\sin y^3+2y\sin x}$ $\dfrac{dy}{dx}=\dfrac{-y\cos x}{3y\sin y^3+2\sin x}$ $\dfrac{dx}{dy}=\dfrac{-(3y\sin y^3+2\sin x)}{y\cos x}$			
	(b)	(i)	$x=e^{-2t}$ $\qquad$ $y=te^{-t}$ $\dfrac{dx}{dt}=-2e^{-2t}$	K1	Find $\dfrac{dx}{dt}$
			$\dfrac{dy}{dt}=e^{-t}+t(-e^{-t})=e^{-t}(1-t)$	K1	Find $\dfrac{dy}{dt}$
			$\dfrac{dy}{dx}=e^{-t}(1-t)\times\dfrac{1}{-2e^{-2t}}$ $=-\dfrac{e^t}{2}(1-t)$	K1√	Find $\dfrac{dy}{dx}$
			$\dfrac{dy}{dx}=0$ $-\dfrac{e^t}{2}(1-t)=0$	K1√	Attempt to find stationary point
			$-\dfrac{e^t}{2}\neq 0,\ t=1$	J1	
			when $t=1$ $x=e^{-2t}=e^{-2}$ $y=te^{-t}=e^{-1}$	K1√	Find x and y
			Stationary point = (e^{-2},e^{-1})	J1	
		(ii)	$\dfrac{d^2y}{dx^2}=[(1-t)(-\dfrac{e^t}{2})+(-\dfrac{e^t}{2})(-1)]\left[\dfrac{1}{-2e^{-2t}}\right]$ $=-\dfrac{e^t}{2}[(1-t)-1]\left[\dfrac{1}{-2e^{-2t}}\right]$ $=-\dfrac{te^{3t}}{4}$	K1√	Correct formula and multiply with $\frac{dt}{dx}$
			when $t=1$ $\dfrac{d^2y}{dx^2}=-\dfrac{e^3}{4}<0$	K1	Subs his value t
			Stationary point = (e^{-2},e^{-1}) is maximum point	J1	State
			TOTAL	**16 M**	

7.	(a)	V = Vol Cylinder − Vol of two hemisphere $= \pi x^2 h - \frac{4}{3}\pi x^3$ $= 40\pi x^2 - \frac{4}{3}\pi x^3$	**B1** **K1J1**	
	(b)	$\frac{dv}{dx} = 80\pi x - 4\pi x^2$ when $\frac{dv}{dx} = 0$ $80\pi x - 4\pi x^2 = 0$ $4\pi x(20 - x) = 0$ $x = 0, \quad x = 20$ $\frac{d^2V}{dx^2} = 80\pi - 8\pi x$ when $x = 0$, $\frac{d^2V}{dx^2} = 80\pi - 8\pi(0) = 80\pi > 0$ when $x = 20$, $\frac{d^2V}{dx^2} = 80\pi - 8\pi(20) = -80\pi < 0$ Volume maximum when $x = 20$ Maximum volume = $40\pi(20)^2 - \frac{4}{3}\pi(20)^3$ $= \frac{16000\pi}{3}$ cm^3	**K1**√ **J1** **K1**√ **J1** **K1**√ **J1**	Differentiate correctly & $\frac{dv}{dx} = 0$ Find max volume
		TOTAL	**9 M**	